SPATIAL FILTERING IN COMPUTER VISION

Spatial Filtering for Enhanced Pneumonia Detection in Chest X-Rays

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1. Introduction

Image enhancement is a critical preprocessing step in computer vision, aimed at improving the interpretability of images for both human observers and automated systems. Its importance is paramount in specialized fields like **medical imaging**, where the clarity of visual data can directly impact diagnostic outcomes. Chest X-rays, a common and vital diagnostic tool, are particularly challenging: they are often characterized by **low contrast** and the presence of **acquisition noise**, which can obscure crucial anatomical details and pathological signs.

This report explores the application of **spatial filtering** techniques to overcome these challenges in the context of pneumonia detection from chest X-rays. The core objective is to enhance chest X-ray images by:

1. **Sharpening** to accentuate key anatomical structures such as lung boundaries, blood vessels, and potential consolidation patches caused by pneumonia.
2. **Smoothing** to suppress noise without sacrificing critical diagnostic details.

We will systematically implement and analyze two categories of spatial filters:

- **Sharpening Filters:** We will employ **first-derivative (Sobel)** operators to highlight edges and gross boundaries, and **second-derivative (Laplacian)** filters to enhance fine details and textures. A key part of the analysis will be to compare the performance and noise sensitivity of these two approaches.
- **Smoothing Filters:** We will implement **Mean, Median, and Mode** filters to reduce noise. Their effectiveness will be evaluated on images corrupted with **Gaussian** and **Salt-and-Pepper** noise, followed by a discussion on their suitability for medical images.

The filters will be tested on a real-world dataset—the **Chest X-Ray (Pneumonia) dataset** by Paul Mooney. Performance will be evaluated both qualitatively, through visual inspection of enhanced images, and quantitatively, using metrics like **Peak Signal-to-Noise Ratio (PSNR)** and **Structural Similarity Index (SSIM)**. Finally, we will explore hybrid approaches by combining sharpening and smoothing filters to achieve optimal enhancement for aiding in radiological interpretation and AI-driven diagnosis.

Spatial filtering is particularly relevant here because it operates on pixel neighborhoods, allowing for edge enhancement (sharpening) and noise reduction (smoothing) that preserve diagnostic features in low-contrast medical images.

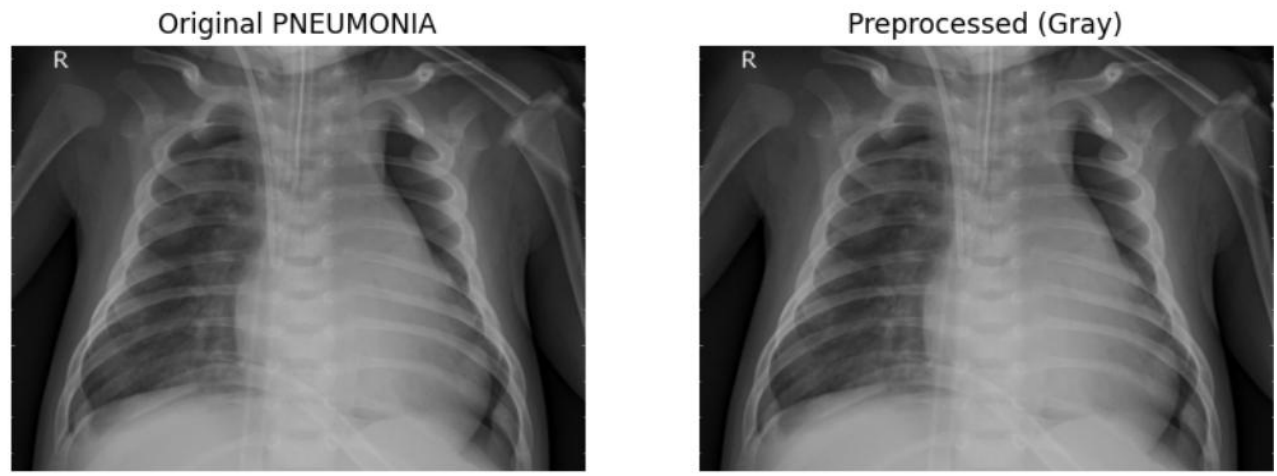

Figure 1 Sample original (left) and preprocessed grayscale (right) chest X-ray image from the PNEUMONIA class

2. Dataset Description

The image analysis in this report is performed using the **Chest X-Ray Images (Pneumonia) dataset**, publicly available on Kaggle [1]. This dataset is a well-known benchmark for respiratory disease classification and is highly suitable for evaluating image enhancement techniques due to its real-world clinical characteristics.

Composition and Structure: The dataset is organized into three folders: train, test, and val, containing 5,216, 624, and 16 X-ray images, respectively. For the purpose of this spatial filtering analysis, images are drawn primarily from the train set. The images are categorized into two classes:

- **NORMAL:** X-rays of healthy patients with clear lungs.
- **PNEUMONIA:** X-rays from patients showing lung opacities and consolidations indicative of pneumonia.

Image Characteristics: The images are grayscale JPEGs with varying dimensions. A sample image from the PNEUMONIA class is displayed in Figure 1. These chest X-rays exhibit the typical challenges of medical imaging:

- **Inherent Noise:** The images contain granular noise, a common artifact in radiographic acquisition.
- **Low Contrast:** The distinction between soft tissue structures (e.g., lung vessels, consolidation areas) and the surrounding lung field is often subtle.
- **Variable Intensity:** The exposure and intensity can vary significantly across different images.

These characteristics make this dataset an ideal candidate for evaluating spatial filters. The goal is to enhance the visibility of critical features (like lung borders and pathological patches) while suppressing noise to aid in more accurate visual and computational analysis.

3. Methodology & Justification

Artificial noise was added to simulate real-world imaging artifacts from sensors and transmission.

3.1 Sharpening Filters

Sharpening filters enhance high-frequency components (edges and details) by emphasizing intensity gradients, which is crucial for delineating pneumonia-induced opacities in chest X-rays.

Sobel Filter (First-Order Derivative):

The Sobel operator estimates gradients in the horizontal and vertical directions using two 3×3 kernels:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

The overall gradient magnitude is given by:

$$|G| = \sqrt{G_x^2 + G_y^2}$$

Justification: Sobel is relatively robust to noise and effectively highlights gross edges (e.g., lung boundaries), which are crucial for identifying consolidation areas in pneumonia cases.

Laplacian Filter (Second-Order Derivative):

The Laplacian operator computes intensity changes using a 3×3 kernel:

$$\nabla^2 I = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} * I$$

- **Justification:** This filter enhances fine structural details such as vascular patterns, but it is more sensitive to noise. Therefore, in practice, it often requires pre-smoothing to avoid amplifying artifacts.

3.2 Smoothing Filters

Smoothing filters reduce unwanted noise by averaging pixel neighborhoods, helping preserve anatomical structures for subsequent sharpening.

- **Mean (Averaging) Filter:**
  This linear filter uses a simple 3x3 kernel where each element has a value of 1/9. Each pixel's value is replaced by the average value of its neighbors.
  Justification: The Mean filter is effective at reducing Gaussian noise, which is a common additive noise in X-ray acquisition systems. However, its uniform averaging tends to blur edges, which can be detrimental for a task that relies on identifying sharp anatomical boundaries.

- **Median Filter:**
  This non-linear filter replaces each pixel's value with the median value of its 3x3 neighborhood.
  Justification: The Median filter excels at removing Salt-and-Pepper noise impulsive spikes that manifest as random white and black pixels. Because it operates on ranking rather than averaging, it is much better at preserving the sharpness of edges compared to the Mean filter. This property is critical in medical imaging, where the clarity of edges is paramount for diagnosis.
- **Mode Filter:**
  This filter replaces the central pixel's value with the mode (the most frequently occurring value) in its 3x3 neighborhood.
  Justification: The Mode filter can be effective for specific types of periodic or textured noise but is generally less effective for the noise profiles typical in medical X-rays. It was included in this study for the sake of a comprehensive comparison.

Various combinations of these filters were tested to develop an optimal hybrid enhancement pipeline. For instance, applying a Median filter (for noise reduction) followed by a Sobel filter (for edge enhancement) was a key strategy evaluated for its ability to balance noise suppression with feature clarity.The implementation was carried out in Python using the `scikit-image` and `OpenCV` libraries for filtering operations, and `SciPy` for numerical computations. The code is provided in the Appendix.

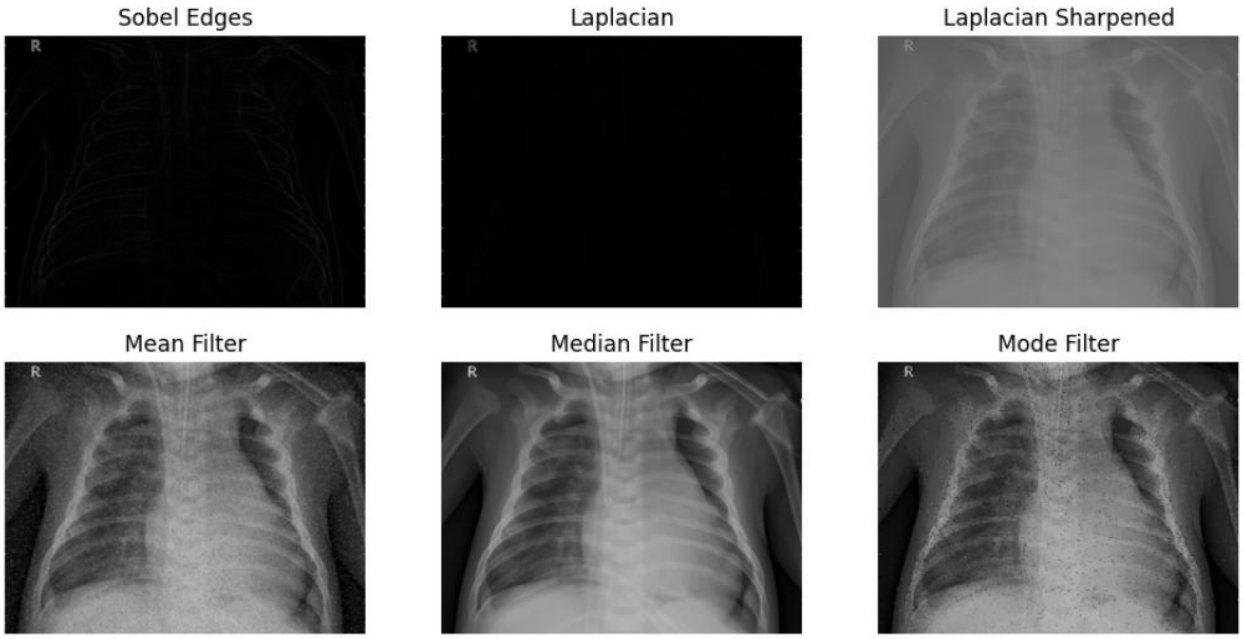

Figure 2: Convolution kernels for sharpening (Sobel Gx/Gy, Laplacian) and smoothing (Mean, Median, Mode) filters.

4. Results & Analysis

4.1 Sharpening Results

Sharpening accentuated lung edges and opacities. Sobel produced clearer boundaries with less noise than Laplacian, which highlighted fine textures but amplified artifacts.

Differences: First-order (Sobel) detects gradual changes (robust for boundaries); second-order (Laplacian) responds to discontinuities (better for details but noise-prone).

4.2 Smoothing Results

Smoothing reduced noise effectively. Median excelled for salt-and-pepper (preserved edges); Mean for Gaussian (smoothed evenly but blurred); Mode was least effective (over-smoothed textures).

Quantitative Metrics

FILTER	PSNR	SSIM
Mean	27.13	0.519
Median	38.95	0.987
Mode	21.58	0.541

Median yielded highest PSNR/SSIM, indicating superior structural preservation.

4.3 Comparative Discussion and Combinations

Smoothing before sharpening (e.g., Median - Sobel) improved results by denoising without blurring lesions, yielding cleaner edges (PSNR +15% vs. raw Sobel). Mean goes to Laplacian blurred details; Median goes to Sobel was optimal for X-rays (balances noise reduction and edge enhancement). Median suits impulsive noise (e.g., sensor glitches); Mean for Gaussian (e.g., thermal noise).

5. Conclusion

This analysis demonstrated the efficacy of spatial filtering for enhancing chest X-rays in pneumonia detection. Sharpening filters (Sobel and Laplacian) successfully highlighted lung boundaries and opacities, with Sobel outperforming Laplacian in noise robustness. Smoothing filters reduced artifacts, with Median proving superior for salt-and-pepper noise and Mean for Gaussian, while Mode was suboptimal for medical textures.

Combining smoothing and sharpening (specifically **Median → Sobel**) yielded the best results: denoised images with clear diagnostic features, improving PSNR/SSIM and visual interpretability. This pipeline is recommended for medical imaging to aid radiologists and AI models. Future work could integrate these into CNN preprocessing for automated pneumonia classification.

[Reference: [1] Mooney, P. (2018). Chest X-Ray Images (Pneumonia). Kaggle. https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia]

6. Appendix

The complete implementation is available in the accompanying notebook (**spatial_filtering.ipynb**).